Data Models

- ☆ Underlying structure of a database.
- ☆ Conceptual tool.
- ☆ Data + Relationship + Semantics + Constraints.
- ☆ Design the database at physical, logical and view level.

Categories of Data Models

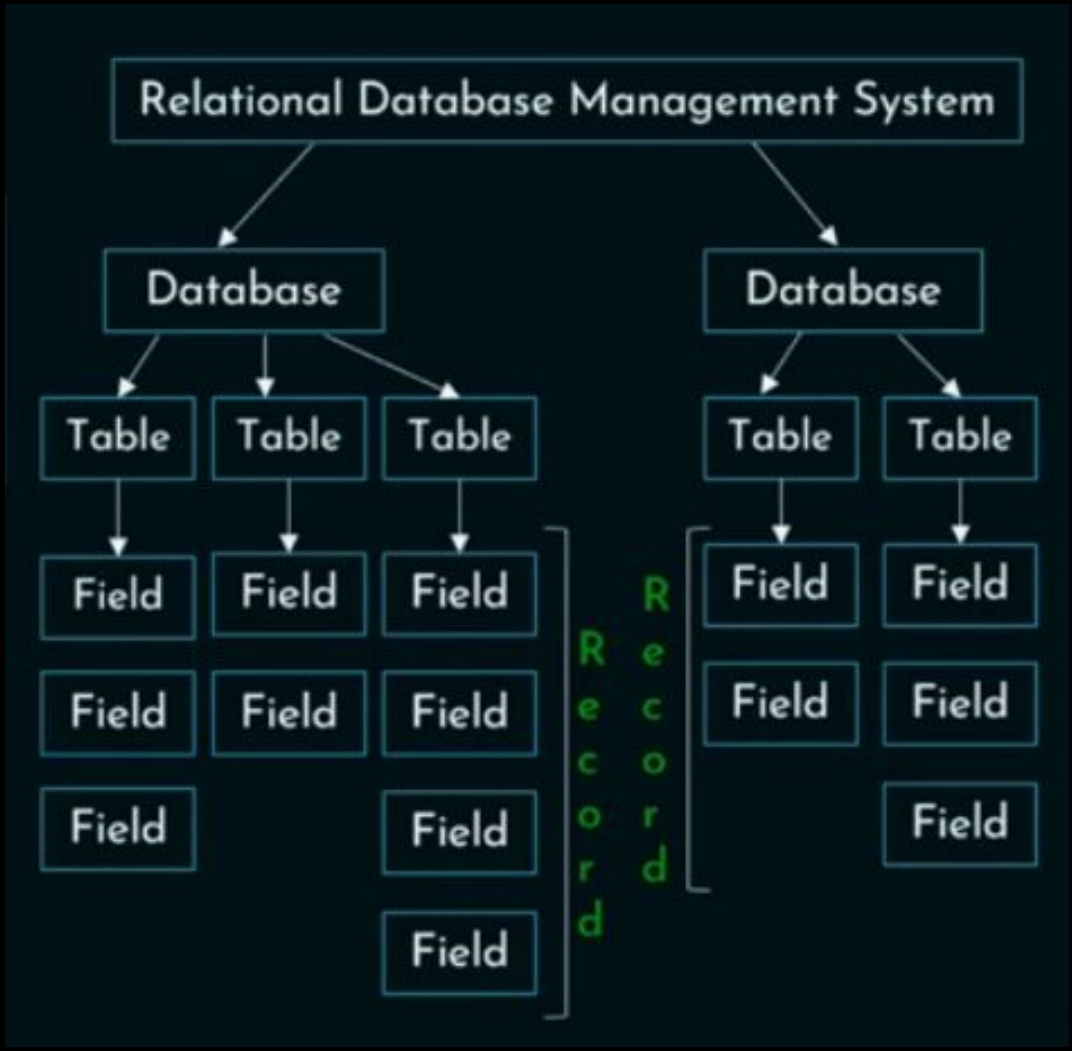
1. Relational Model.
2. Entity-Relationship Model.
3. Object-based Data Model.
4. Semistructured Data Model.

1. Relation al Model

- ☆ Collection of tables.
- ☆ Tables represent both data and relationships.
- ☆ Multiple columns with unique names.
- ☆ Tables = Relations.

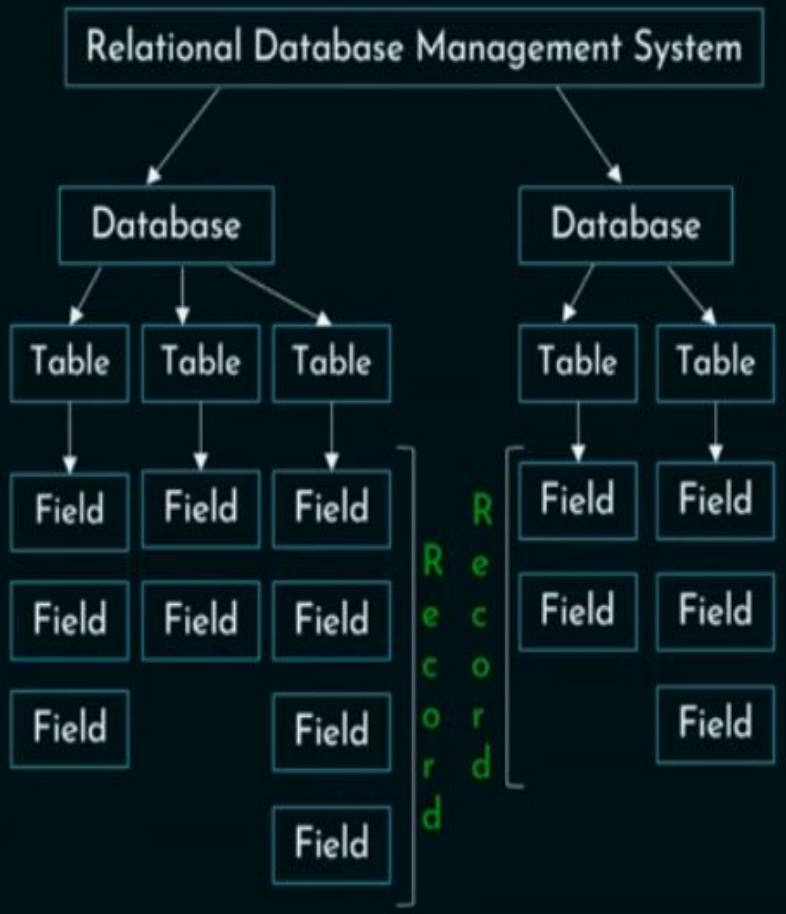
- ☆ Record-based model.
- ☆ Fixed-format records of several types.
- ☆ Each record type defines a fixed number of fields, or attributes.
- ☆ Most widely used data model.
- ☆ Current DBS are based on relational model.

1. Relation al Model



1. Relation al Model

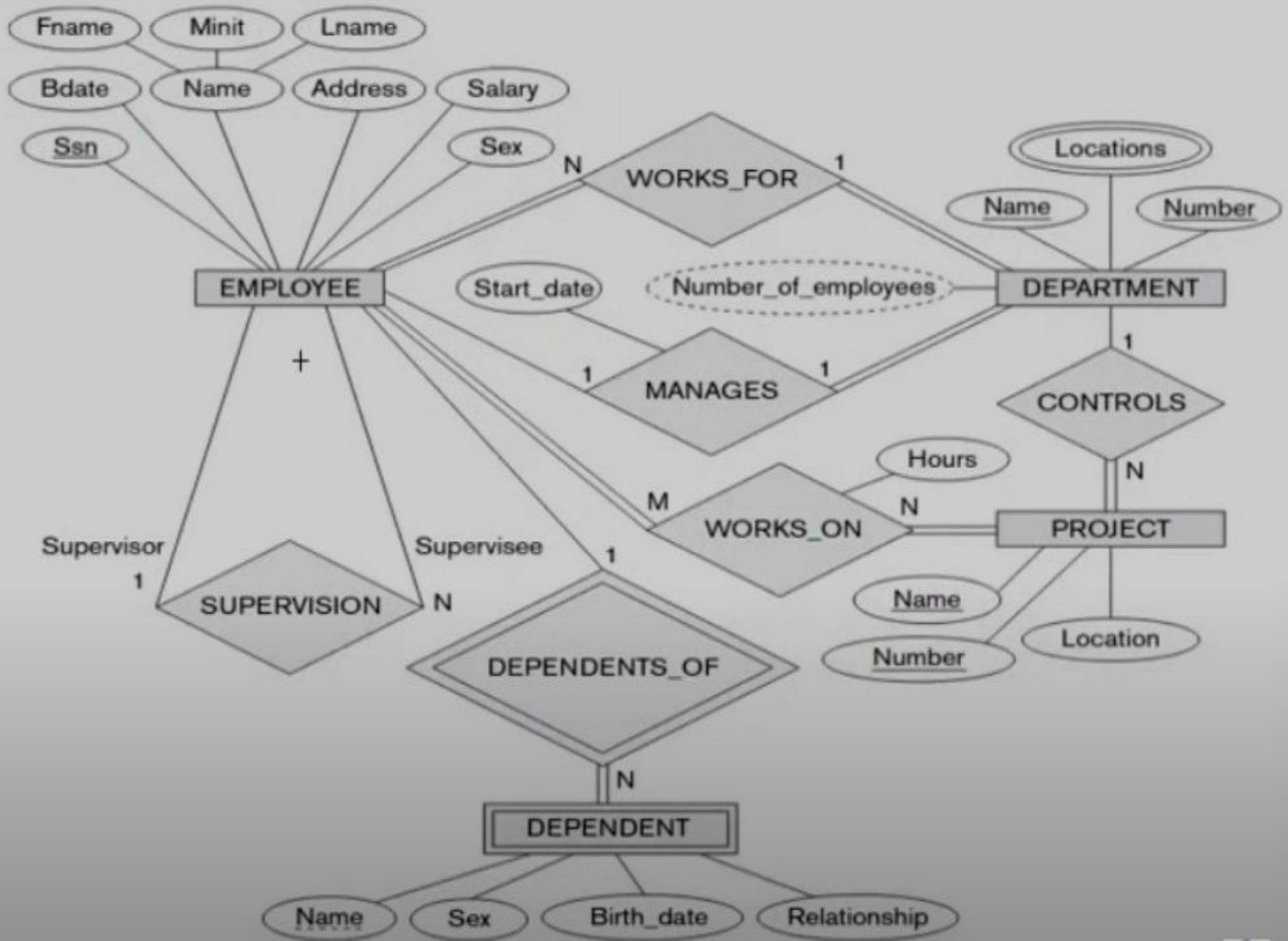
EMPLOYEE	FName	LName	Salary	DNo
	Tom	Ford	30000	3
	Amy	Jones	30000	2



2. Entity Relationship Model

- ☆ E-R data model or E-R model.
- ☆ Collection of basic objects called entities and its relationship.
- ☆ An entity is a real world thing or object.
- ☆ Distinguishable from other objects.
- ☆ Widely used in database design.

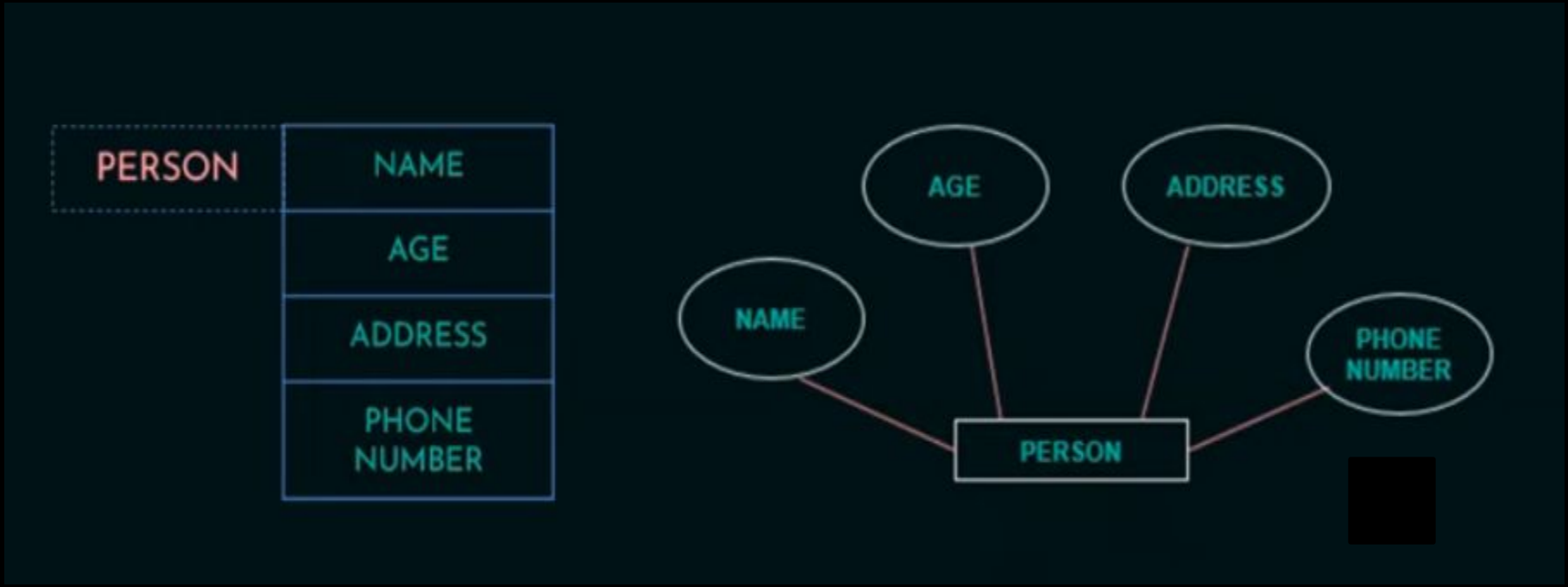
The ER conceptual schema diagram for the COMPANY database.



3. Object based Model

- ☆ OOP Languages - C++, Java, C# etc.,
- ☆ Need for Object-oriented data model.
- ☆ Object-Based Model = E-R Model + OO features.
- ☆ Object-Relational Data Model = Object-Based Model + Relational Model.

3. Object based Model



4. Semistructured Data Model

- ☆ Specification of data.
- ☆ Individual data items of the same type may have different sets of attributes.
- ☆ Contrast to previous data models.
- ☆ Extensible Markup Language (XML).

4. Semistructured Data Model

Example:

```
<Note>
    <From> Neso Academy </From>
    <To> All Students </To>
    <Subject> Thank you </Subject>
    <Body> Thank you for subscribing Neso Academy. Good Luck.
    </Body>
</Note>
```